

Single generators of polynomial rings under composition: a characteristic-two dichotomy

— condensed version —

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Abstract

Say that $p \in R[x, y]$ *generates* $R[x, y]$ if every bivariate polynomial over R is an iterated composition of p , the variables, and constants. We prove: there is a single polynomial generating all of $R[x, y]$ under composition if and only if 2 is a unit of R ; when it is, $x^2 - y$ is a generator, and when it is not, for every p at least one of $x + y$, xy is inexpressible.

1 Definitions and statements

Definition 1. Let R be a commutative ring, $R[x, y]$ the ring of formal polynomials in two variables, and $p \in R[x, y]$. The set $\text{Clo}(p) \subseteq R[x, y]$ is the smallest set containing x , y , and every constant $c \in R$, and closed under $(f, g) \mapsto p(f, g)$ (substitution of f for x and g for y in p). Say p *generates* $R[x, y]$ if $\text{Clo}(p) = R[x, y]$.

In clone-theoretic terms: the binary terms in the language of commutative rings with constant symbols form the binary part of an abstract clone, canonically identified with $R[x, y]$ under substitution, and our question asks when the single term p generates it. This abstract object is represented faithfully by operations on R when R is an infinite integral domain, and is finer than its image otherwise; it should not be confused with the clone of polynomial operations on R in the sense of universal algebra, nor with the full clone of all finitary operations on R . The distinction between formal polynomials and the functions they induce matters over finite fields. For example, $n := 1 + xy \in \mathbb{F}_2[x, y]$ induces the binary NAND operation on $\mathbb{F}_2$ (the Sheffer stroke [2]), and every Boolean function is a composition of NAND; nevertheless $x + y \notin \text{Clo}(n)$ by Theorem 4, since $Dn = 1 \neq 0$ for the operator D of Section 3.

Theorem 2 (dichotomy). *Let R be a commutative ring. There is a single polynomial generating all of $R[x, y]$ under composition (with variables and constants) if and only if 2 is a unit of R .*

Theorem 3. *If $\frac{1}{2} \in R$ then $q(x, y) := x^2 - y$ generates the full polynomial ring: $\text{Clo}(x^2 - y) = R[x, y]$.*

Theorem 4. *For every $q \in \mathbb{F}_2[x, y]$, $x + y \notin \text{Clo}(q)$ or $xy \notin \text{Clo}(q)$.*

There is no hypothesis on q : reduction mod 2 collapses degree ($2x^3 + xy \mapsto xy$), so the deduction of Corollary 6 below requires Theorem 4 for arbitrary q , degenerate cases included.

Theorem 5. *For every field k of characteristic 2 and every $q \in k[x, y]$, at least one of $x + y$, xy lies outside $\text{Clo}(q)$.*

Corollary 6. *No binary polynomial generates $\mathbb{Z}[x, y]$.*

Lemma 7. $\text{Clo}(p) = R[x, y]$ if and only if $x + y \in \text{Clo}(p)$ and $xy \in \text{Clo}(p)$.

Proof. Substituting members of $\text{Clo}(p)$ into a member lands in $\text{Clo}(p)$ (induction on the formula). So if $x + y, xy$ are members, $\text{Clo}(p)$ is closed under $+$ and $\times$; containing the variables and constants, it is everything. $\square$

Lemma 8 (transfer). *A ring homomorphism $R \rightarrow S$ induces $\pi : R[x, y] \rightarrow S[x, y]$ with $\pi(\text{Clo}(p)) \subseteq \text{Clo}(\pi p)$.*

Proof. π fixes variables, sends constants to constants, and commutes with substitution; induct over the formula. $\square$

By Lemmas 7 and 8 applied to $\mathbb{Z} \rightarrow \mathbb{F}_2$, Theorem 4 implies Corollary 6.

We use three facts about the formal partials ∂_x, ∂_y on $\mathbb{F}_2[x, y]$: (1) $\partial(f^2) = 2f \partial f = 0$; (2) $\partial_x^2 = \partial_y^2 = 0$, while $D := \partial_x \partial_y$ need not vanish; (3) squaring is a ring homomorphism, and a polynomial with all exponents even is a square. Univariate degrees follow the convention that constants — including 0 — have degree 0, so $\deg(f^2) = 2 \deg f$ without exception.

2 The positive half: $x^2 - y$

Proof of Theorem 3. By Lemma 7 it suffices to reach $x + y$ and xy . All steps below are instances of

$$q(\alpha, q(\alpha', \gamma)) = \alpha^2 - \alpha'^2 + \gamma. \quad (1)$$

Translations. With constants $\alpha = \frac{d+1}{2}$, $\alpha' = \frac{d-1}{2}$, (1) sends $\gamma \mapsto \gamma + d$ for any $d \in R$ (the only step using $\frac{1}{2}$); in particular $x + d, y + d \in \text{Clo}(q)$.

Slanted lines. $q(c, y) = c^2 - y$, then $q(x, c^2 - y) = x^2 + y - c^2$, then

$$q(x + d, x^2 + y - c^2) = 2dx - y + (d^2 + c^2) \in \text{Clo}(q),$$

a line of slope $2d$ for every d , constant adjustable by translations.

Linear engine. For $\alpha \in \text{Clo}(q)$ and constant k , (1) with outer argument $\alpha + k$ and inner argument α gives the increment

$$\gamma \mapsto \gamma + 2k\alpha + k^2.$$

Increments along slanted lines of two distinct slopes span all linear polynomials, so from $\gamma = 0$: every linear polynomial lies in $\text{Clo}(q)$, in particular $x + y$ and $x + \frac{y}{2}$.

Scaled squares. The engine with $\alpha = y^2 + \ell$ (in the clone via $q(y, -\ell)$) adds $\lambda y^2 + \text{const}$ for any λ ; from $\gamma = x^2 = q(x, 0)$ we get $x^2 + \frac{1}{4}y^2 \in \text{Clo}(q)$.

Finish.

$$q\left(x + \frac{y}{2}, x^2 + \frac{1}{4}y^2\right) = \left(x + \frac{y}{2}\right)^2 - x^2 - \frac{1}{4}y^2 = xy. \quad \square$$

3 The negative half over $\mathbb{F}_2$

Fix $q \in \mathbb{F}_2[x, y]$. Grouping the monomials of q by the parities of their exponents and using fact (3), q has a unique *parity decomposition*

$$q = A_0^2 + xA_1^2 + yA_2^2 + xyA_3^2, \quad A_i \in \mathbb{F}_2[x, y], \quad (2)$$

and differentiating, with squares derivative-free,

$$\partial_x q = A_1^2 + yA_3^2, \quad \partial_y q = A_2^2 + xA_3^2, \quad Dq = A_3^2. \quad (3)$$

Lemma 9 (chain rule for D). *In characteristic 2, for all $q, \alpha, \beta \in \mathbb{F}_2[x, y]$,*

$$D(q(\alpha, \beta)) = (Dq)(\alpha, \beta) (\partial_x \alpha \partial_y \beta + \partial_y \alpha \partial_x \beta) + (\partial_x q)(\alpha, \beta) D\alpha + (\partial_y q)(\alpha, \beta) D\beta.$$

Proof. Expand $\partial_x \partial_y (q(\alpha, \beta))$ by the chain and product rules; the terms involving $\partial_x^2 q$ or $\partial_y^2 q$ vanish by fact (2), and the survivors assemble as displayed. $\square$

Proposition 10. *If $Dq = 0$ then $xy \notin \text{Clo}(q)$.*

Proof. $\{g : Dg = 0\}$ contains the variables and constants and, by Lemma 9 with the first term dead, is closed under substitution into q ; hence it contains $\text{Clo}(q)$. But $D(xy) = 1$. $\square$

Suppose henceforth $Dq \neq 0$, i.e. $A_3 \neq 0$; we show $x + y \notin \text{Clo}(q)$. Write $\mathbb{F}_2[x+y]$ for the subring of polynomials in $x + y$. The proof runs through the rigidity property

($\dagger$) for all $\alpha, \beta \in \mathbb{F}_2[x, y]$: whenever $q(\alpha, \beta)$ is a nonconstant element of $\mathbb{F}_2[x+y]$, both $\alpha, \beta \in \mathbb{F}_2[x+y]$.

Proposition 11 (master reduction). *If q satisfies ($\dagger$) then $x + y \notin \text{Clo}(q)$.*

Proof. We claim every $g \in \text{Clo}(q)$ satisfies $P(g) : g \notin \mathbb{F}_2[x+y]$ or g constant. The substitution $x \leftrightarrow y$ fixes $\mathbb{F}_2[x+y]$ pointwise but moves x and y , so the variables satisfy P ; constants do. Let $g = q(\alpha, \beta)$ with $P(\alpha), P(\beta)$ and suppose $g \in \mathbb{F}_2[x+y]$ is nonconstant. Then ($\dagger$) forces $\alpha, \beta \in \mathbb{F}_2[x+y]$; P forces them constant; then g is constant — contradiction. So P holds on $\text{Clo}(q)$, and $x + y$ fails P . $\square$

Fix an algebraically closed field $K \supseteq \mathbb{F}_2$ of characteristic 2 containing an element σ transcendental over $\mathbb{F}_2$; concretely $K = \overline{\mathbb{F}_2(\sigma)}$. Call $c \in K$ *transcendental* if it satisfies no nonzero polynomial relation over $\mathbb{F}_2$.

Definition 12. A *witness* is a pair $\alpha, \beta \in K[t]$, not both constant, with $q(\alpha, \beta) = c$ constant and c transcendental.

Proposition 13 (bridge). *If no witness exists, then q satisfies ($\dagger$).*

Proof. Let $q(\alpha, \beta) = f(x + y)$ with $f \in \mathbb{F}_2[u]$ nonconstant. Substitute $(x, y) = (t, t + \sigma)$: the right side becomes the constant $f(\sigma)$, which is transcendental (a nonzero $\mathbb{F}_2$ -relation on $f(\sigma)$ composes with f to one on σ). Since no witness exists, $\alpha(t, t + \sigma)$ and $\beta(t, t + \sigma)$ are constant in t . Finally, a $g \in \mathbb{F}_2[x, y]$ constant along the generic line lies in $\mathbb{F}_2[x+y]$: expanding $g(x, x + u) = \sum_i g_i(u) x^i$, constancy gives $g_i(\sigma) = 0$ for $i \geq 1$, hence $g_i = 0$, so $g = g_0(x + y)$. $\square$

It remains to prove:

Theorem 14 (no witnesses). *If $Dq \neq 0$, no witness exists.*

4 The Frobenius descent

Throughout, $'$ denotes d/dt and degrees are degrees in t . Over $K[t]$: squares have zero derivative, and conversely a polynomial with zero derivative is a square (it is a polynomial in t^2 , and the coefficients have square roots since K is perfect).

Set $w := \gcd(A_1, A_2, A_3)$ and $A_i = wB_i$, so the B_i have no common nonunit factor and

$$q = A_0^2 + w^2 h, \quad h := xB_1^2 + yB_2^2 + xyB_3^2, \quad (4)$$

with $B_3 \neq 0$ and partials $h_x = B_1^2 + yB_3^2$, $h_y = B_2^2 + xB_3^2$, $Dh = B_3^2$. The peel (4) removes the possible curve of critical points of q along $\{w = 0\}$; Lemma 17 shows none survives in h . Three lemmas, then the descent.

Lemma 15 (kill lemma). *Let (α, β) be a witness with level c , and $f \in \mathbb{F}_2[x, y]$, $f \neq 0$. Then $f(\alpha, \beta) \neq 0$ in $K[t]$.*

Proof sketch. Suppose $f(\alpha, \beta) = 0$; after swapping variables assume α nonconstant. View f and $g := q + z$ as polynomials in y over $\mathbb{F}_2[x, z]$: since g is monic and linear in z it is irreducible and cannot divide the z -free f , so they are coprime, and a Bézout identity with denominators cleared gives $fU + (q + z)V = E(x, z) \neq 0$ in $\mathbb{F}_2[x, z][y]$. Substituting $y \mapsto \beta$, $x \mapsto \alpha$, $z \mapsto c$ kills both summands ($q(\alpha, \beta) + c = 0$), so $E(\alpha, c) = 0$; expanding $E = \sum_i \rho_i(z) x^i$ and using that $1, \alpha, \alpha^2, \dots$ have strictly increasing degrees in t , every $\rho_i(c) = 0$, and some $\rho_i \neq 0$ exhibits c as algebraic over $\mathbb{F}_2$ — contradiction. $\square$

Lemma 16 (algebraic points). *If $u, v \in \mathbb{F}_2[x, y]$ have no common nonunit factor and $(a, b) \in K^2$ satisfies $u(a, b) = v(a, b) = 0$, then a, b are algebraic over $\mathbb{F}_2$.*

Proof sketch. In $\mathbb{F}_2(x)[y]$ a common divisor would descend to $\mathbb{F}_2[x, y]$ by Gauss's lemma [1, Ch. IV], so u, v are coprime there; a cleared Bézout identity gives $su + tv = r(x) \neq 0$, and evaluation at (a, b) gives $r(a) = 0$. Symmetrically for b . $\square$

Lemma 17 (keystone). *Let $B_1, B_2, B_3 \in \mathbb{F}_2[x, y]$ have no common nonunit factor, with $B_3 \neq 0$, and set $h_x := B_1^2 + yB_3^2$, $h_y := B_2^2 + xB_3^2$. Then h_x and h_y have no common nonunit factor.*

Proof. Both are nonzero: applying ∂_y to a relation $h_x = 0$ would give $B_3^2 = 0$. Suppose a prime p divides both.

If $p \mid B_3$: subtracting the B_3^2 -terms, p divides B_1^2 and B_2^2 , hence (primality) all three B_i — excluded.

If $p \nmid B_3$: write $h_x = pu$ and $h_y = pv$. Differentiate, remembering squares are derivative-free:

$$0 = \partial_x h_x = p_x u + p u_x \implies p \mid p_x u;$$

$$B_3^2 = \partial_y h_x = p_y u + p u_y \implies p \nmid u$$

(else $p \mid B_3^2$, so $p \mid B_3$);

$$B_3^2 = \partial_x h_y = p_x v + p v_x \implies p \nmid p_x$$

(same reason). So the prime p divides $p_x u$ but neither factor — contradiction. $\square$

Proof of Theorem 14. Among all witnesses take one minimizing $n := \deg \alpha + \deg \beta$ (≥ 1 , as not both are constant); fix $r := \sqrt{c}$, transcendental (if r were algebraic, so were $c = r^2$). By Lemma 15 applied to the nonzero polynomials w, h_x, h_y, B_3 , none of

$$W := w(\alpha, \beta), \quad A := h_x(\alpha, \beta), \quad B := h_y(\alpha, \beta), \quad \Delta := B_3(\alpha, \beta)^2$$

vanishes in $K[t]$.

Peel. From (4) and $q(\alpha, \beta) = c = r^2$, using $(u + v)^2 = u^2 + v^2$,

$$W^2 H = S^2, \quad H := h(\alpha, \beta), \quad S := r + A_0(\alpha, \beta); \quad (5)$$

differentiating (W^2 and S^2 are squares) gives $W^2 H' = 0$, so $H' = 0$.

Engine. The chain rule for h , with $\partial_x h_x = \partial_y h_y = 0$ and $\partial_y h_x = \partial_x h_y = Dh = B_3^2$, gives

$$A\alpha' + B\beta' = H' = 0, \quad A' = \Delta\beta', \quad B' = \Delta\alpha',$$

whence $(AB)' = \Delta(B\beta' + A\alpha') = 0$ and AB is a square in $K[t]$.

Fork, branch 1: A and B share a root $\theta \in K$. Then $P := (\alpha(\theta), \beta(\theta))$ is a common zero of h_x, h_y , which by Lemmas 17 and 16 has algebraic coordinates. Evaluating (5) at θ , $S(\theta)^2 = w(P)^2 h(P)$ is algebraic, hence so is $S(\theta)$, hence so is $r = S(\theta) + A_0(P)$ — contradicting transcendence of r .

Fork, branch 2: A, B coprime. Coprime factors of a square are squares (units of $K[t]$ are constants of the algebraically closed K , hence squares), so $A' = B' = 0$; since $\Delta \neq 0$, the engine identities force $\alpha' = \beta' = 0$, so $\alpha = \alpha_1^2, \beta = \beta_1^2$ with $\deg \alpha_1 + \deg \beta_1 = n/2$. Because the coefficients of q lie in $\mathbb{F}_2$, squaring commutes with substitution into q :

$$q(\alpha_1, \beta_1)^2 = q(\alpha_1^2, \beta_1^2) = c,$$

so $q(\alpha_1, \beta_1) = r$ and (α_1, β_1) is a witness of total degree $n/2 < n$ — contradicting minimality. (If $n = 1$ halving is impossible and this branch cannot occur, so the descent is well-founded.) $\square$

Theorem 14 gives $(\dagger)$ by Proposition 13, hence $x + y \notin \text{Clo}(q)$ by Proposition 11 when $Dq \neq 0$; with Proposition 10 this proves Theorem 4, and with Lemma 8, Corollary 6.

5 From $\mathbb{F}_2$ to the dichotomy

Proposition 18. *For every perfect field k of characteristic 2 and every $q \in k[x, y]$, at least one of $x + y, xy$ lies outside $\text{Clo}(q)$.*

Proof. The descent runs over k with one change: coefficients of q are no longer fixed by squaring, so in the halving branch $q(\alpha_1^2, \beta_1^2) = q^{(1/2)}(\alpha_1, \beta_1)^2$, where $q^{(1/2)}$ applies the inverse Frobenius to each coefficient (perfectness enters here); the halved pair is a witness for $q^{(1/2)}$, and since the induction is on witness degree with q universally quantified inside, the twist is absorbed. Parity decomposition, bridge, and master reduction are unaffected. $\square$

Proof of Theorem 5. k embeds into its perfection $k^{1/2^\infty}$; by Lemma 8, membership of both $x + y$ and xy in $\text{Clo}(q)$ over k would persist over the perfection, contradicting Proposition 18. $\square$

Proof of Theorem 2. If 2 is a unit, $x^2 - y$ generates $R[x, y]$ (Theorem 3). If not, 2 lies in a maximal ideal $\mathfrak{m}$; a generator would push forward along $R \rightarrow R/\mathfrak{m}$ (Lemma 8) to one over the residue field, of characteristic 2 — contradicting Theorem 5. $\square$

All results have been machine-verified in Lean 4 against Mathlib (axioms: propositional extensionality, quotient soundness, choice); the development accompanies the paper.

Three questions remain open. Which degrees ≥ 3 admit single generators over $\mathbb{Q}$ ($x^3 - y$ is a candidate; over $\mathbb{F}_3$, $\text{Clo}(x^3 - y)$ consists of additive-plus-constant polynomials, so $x^3 - y$ does not generate there)? Which p generate when 2 is a unit ($xy + 1$ and $x^2 + y$ do not, by a characteristic-zero instance of $(\dagger)$ and by a positivity obstruction, respectively)? Is membership in $\text{Clo}(p)$, or generation, decidable? The identity $q(f, q(f, g)) = g$ for $q = x^2 + y$ over $\mathbb{F}_2$ shows substitution admits unbounded cancellation, so bounded search proves nothing.

References

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